

Title :

AI-Powered Personal Finance Tracker (FinAI / MoneyMentor)

Abstract:

Managing personal finances effectively remains a challenge for many individuals due to the complexity and inconsistency of financial data sources like bank statements, SMS alerts, and transaction summaries. This project proposes an AI-powered personal finance tracker that automates the process of analyzing expenses, categorizing transactions, and generating personalized financial insights.

The system enables users to upload SMS records or bank transaction files (CSV format), which are then processed using Natural Language Processing (NLP) and machine learning techniques to extract and classify expenses into meaningful categories such as food, travel, utilities, and shopping. It further identifies spending patterns, calculates monthly trends, and offers intelligent suggestions to help users reduce unnecessary expenses and improve savings behavior.

An interactive dashboard provides users with visual summaries, category-wise breakdowns, and smart insights into their spending habits. The platform emphasizes ease of use, automation, and personalized recommendations, making it a valuable tool for budgeting and financial awareness.

This project demonstrates the practical application of AI in the domain of personal finance and encourages better money management through intelligent automation and data-driven decision-making.

Keywords:

AI-powered finance, personal expense tracking, transaction categorization, machine learning, natural language processing, financial behavior analysis, budget planning, intelligent recommendations, financial automation, user-centric dashboards.

Key Features:

1. Automatic Transaction Extraction

Extracts expenses and income data from user-uploaded SMS logs or CSV bank statements using NLP techniques.

2. Smart Expense Categorization

Classifies transactions into categories like Food, Transport, Bills, and Shopping using machine learning or rule-based logic.

3. Spending Pattern Analysis

Analyzes monthly trends, detects unusual spending behavior, and highlights major expense areas.

4. Personalized Saving Recommendations

Generates actionable suggestions to help users cut down on non-essential expenses and improve budgeting.

5. Interactive Dashboard

Presents category-wise summaries, pie charts, and time-based graphs for easy tracking and understanding of personal finances.

6. Recurring Payment Detection

Identifies repeated transactions (e.g., rent, subscriptions) to help users stay aware of fixed monthly obligations.

Workflow :

